CHAPTER 3 • PART 1

Variables in C# Console

A complete guide to all 5 variable types — declaration, output, and user input with working code examples.

So today we going to talk about one important topic in whole programming language domain not only restricted in c# that is the **variables**. Consider it like a **box or a container** in which you store food or your own belongings. Now in our programming domain there are **five different types of variables** which are as follows:

- **`int`** 4 bytes
  Integer — Whole numbers like 1, 2, 3 etc.
- **`double`** 8 bytes
  Double — Decimal numbers like 1.55, 2.78, 3.01 etc.
- **`string`** 6 bytes
  String — Words like "soham", "console" etc.
- **`char`** 1 byte
  Char — Single letter like A, c, B etc.
- **`bool`** 2 bytes
  Boolean — True and False, used for condition checking

Now let us explain each variable in detail.

01 Integer Type Variable

This type of variables are usually **whole numbers** like 1, 2, 3 etc. The size of an integer is **4 bytes** not in c# but in the whole programming domain. Now writing a simple code we will show how it is declared.

■ Declare and print:

```csharp
using System; class Program {
  static void Main(string[] args) {
    int x = 12;  Console.WriteLine(x);
  } }
```

At first we declared an integer type variable **`int`** in which we assigned a value 12 — consider 'int' be a container and 12 be the food kept in it. The output will come only **`12`**.

■ With message:

```csharp
    Console.WriteLine("The output is : " + x);  // Output: The output is : 12
```

■ User input (must convert — input is always string type):

```
int x = Convert.ToInt32(Console.ReadLine());  // OR: int.Parse(Console.ReadLine())
Console.WriteLine("The number is : " + x);
```

02 Double Type Variable

This type of variables are usually **decimal numbers** like 1.55, 2.78, 3.01 etc. The size of a double is **8 bytes** not in c# but in the whole programming domain.

■ Declare and print:

```
double x = 12.5; Console.WriteLine(x);  // Output: 12.5
```

At first we declared a decimal type variable **`double`** in which we assigned a value 12.5. The output will come only **`12.5`**.

■ With message:

```
Console.WriteLine("The output is : " + x);  // Output: The output is : 12.5
```

■ User input (must convert):

```
double x = Convert.ToDouble(Console.ReadLine());  // OR: double.Parse(...)
Console.WriteLine("The number is : " + x);
```

03 String Type Variable

This type of variables are usually **words** like soham, console etc. The size of a string is **6 bytes** not in c# but in the whole programming domain.

■ Declare and print:

```
string name = "console"; Console.WriteLine(name);  // Output: console
```

At first we declared a string type variable in which we assigned a value console. The output will come only **`console`**.

■ With message:

```
Console.WriteLine("The name is : " + name);  // Output: The name is : console
```

■ User input (no conversion needed — input is already string type):

```
string name = Console.ReadLine();
Console.WriteLine("The name is : " + name);
```

04 Char Type Variable

This type of variables are usually a **single letter** like A, c, B etc. The size of a char is **1 byte** not in c# but in the whole programming domain. Always use **`single quotes`** for char values.

Declare and print:

```
char letter = 'A'; Console.WriteLine(letter);  // Output: A
```

At first we declared a char type variable in which we assigned value 'A'. The output will come only **A**.

With message:

```
Console.WriteLine("The output is : " + letter);  // Output: The output is : A
```

User input (cast to char):

```
char letter = (char)Console.Read();
Console.WriteLine("The output is : " + letter);
```

05 Boolean Type Variable

This type of variables works on two cases: **True** and **False**. The size of a bool is **2 bytes** not in c# but in the whole programming domain. It is mainly used for **condition checking**.

Declare and print:

```
bool isAlive = true; Console.WriteLine(isAlive);  // Output: True
```

At first we declared a boolean type variable 'isAlive' assigned as true. The output will come only **`True`**.

With message:

```
Console.WriteLine("The output is : " + isAlive);  // Output: The output is : True
```

User input (must convert):

```
bool value = Convert.ToBoolean(Console.ReadLine());  // OR: bool.Parse(...)
Console.WriteLine("The output is : " + value);
```

In the next part we will discuss about how to get the size of variables. So Good bye ■

Quick Reference

TYPE	KEYWORD	SIZE	EXAMPLE	QUOTES
Integer	`int`	4 bytes	12, -5, 100	None
Decimal	`double`	8 bytes	3.14, 9.99	None
Text	`string`	varies	"hello"	Double " "
Char	`char`	1 byte	'A', 'z'	Single ' '
Boolean	`bool`	2 bytes	true, false	None